# AI.104

## Computer Vision

### COURSE OVERVIEW

How machines see: image formation, filtering, feature detection, and introductory object recognition.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

# ORIENTATION: THE MACHINE EYE

**KEY TOPICS**

- Image formation and the camera model
- Pixels, color spaces, and digital images
- The computer vision pipeline
- Vision in robotics and AI

**TEACHING FOCUS**

Understand how images become machine-readable data.

**KEY TAKEAWAY** Seeing begins with light, lenses, and sensors.

2

SESSION

60 MINUTES

# IMAGE PROCESSING & FILTERING

**KEY TOPICS**

- Point operations: brightness, contrast
- Convolution and spatial filtering
- Smoothing: Gaussian and median
- Edge detection: Sobel, Canny

**TEACHING FOCUS**

Enhance and extract structure from images.

▲ **KEY TAKEAWAY** Filters reveal what the eye may miss.

COMPUTER VISION

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SESSION

60 MINUTES

## FEATURES & KEYPOINTS

**KEY TOPICS**

- Corners: Harris and Shi-Tomasi
- SIFT and ORB descriptors
- Feature matching and tracking
- Template matching

**TEACHING FOCUS**

Identify stable points across views.

**KEY TAKEAWAY** Keypoints anchor vision to the world.

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SESSION

60 MINUTES

# IMAGE SEGMENTATION

**KEY TOPICS**

- Thresholding and Otsu
- Region growing and watershed
- Contour detection
- Semantic vs. instance segmentation

**TEACHING FOCUS**

Partition an image into meaningful regions.

**KEY TAKEAWAY** Segmentation separates object from background.

5

SESSION

60 MINUTES

## OBJECT RECOGNITION BASICS

**KEY TOPICS**

- Sliding-window detection
- Histogram of oriented gradients
- Intro to deep vision
- Face and object recognition

**TEACHING FOCUS**

Name and locate things in images.

**KEY TAKEAWAY** Recognition answers 'what is where?'.

COMPUTER VISION

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SESSION

60 MINUTES

## VISION FOR ROBOTS & REVIEW

**KEY TOPICS**

- Visual servoing and motion
- Camera calibration
- Course recap and assessment
- Toward machine vision (AI.307)

**TEACHING FOCUS**

Connect vision to robot action.

▲ **KEY TAKEAWAY** Robots act on what they see.